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INF:3201 INFORMATION SYSTEM AUDIT AND CONTROL

<ASSIGNMENT>

**DEVELOPMENT OF INFORMATION SECURITY POLICY FRAMEWORK FOR
NEXTGEN BUSINESS SOLUTIONS LTD**

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INTRODUCTION

Information has become one of the most valuable assets within modern organizations. Organizations increasingly depend on information systems to manage daily operations, facilitate communication, support decision-making, and maintain competitiveness. However, the growing use of technology has also increased exposure to cybersecurity threats such as unauthorized access, malware attacks, phishing attacks, data breaches, insider threats, and system failures.

To address these challenges, organizations implement information security policies that establish rules, procedures, and responsibilities for protecting information assets. Information security policies ensure that information remains confidential, accurate, and available whenever required.

This report presents an Information Security Policy Framework for NextGen Business Solutions Ltd, a technology company that uses an Enterprise Resource Planning (ERP) system consisting of Human Resource, Finance, Procurement, and Sales modules.

2.0 COMPANY BACKGROUND

NextGen Business Solutions Ltd is a medium-sized technology company operating in Tanzania. The company provides ICT equipment, software solutions, IT consultancy services, and technical support to public and private organizations.

To improve efficiency and integration of business processes, the company uses an Enterprise Resource Planning (ERP) system. The ERP system contains four major modules:

Human Resource Module – manages employee information, recruitment, payroll, attendance, and performance records.

Finance Module – manages accounting activities, budgeting, invoicing, payments, and financial reporting.

Procurement Module – manages purchasing activities, supplier records, inventory management, and procurement approvals.

Sales Module – manages customer information, product sales, customer orders, and sales reporting.

The ERP system stores critical organizational data; therefore, a comprehensive information security framework is required.

3.0 INFORMATION SECURITY POLICY

3.1 Purpose

The purpose of this policy is to establish a framework that protects organizational information assets from unauthorized access, disclosure, modification, destruction, or disruption.

3.2 Scope

This policy applies to:

- Employees
- Contractors
- Consultants

- Vendors
- Third-party service providers
- Information systems
- ERP applications
- Databases
- Computer networks
- Mobile devices

3.3 Objectives

The objectives of information security are:

- Protect information assets.
- Ensure business continuity.
- Maintain customer trust.
- Ensure regulatory compliance.
- Reduce security risks.
- Support secure business operations.

3.4 Information Security Principles

Confidentiality

Information shall only be accessible to authorized users. Examples include employee records, financial statements, customer information, and supplier contracts.

Integrity

Information shall remain accurate, complete, and protected from unauthorized modifications.

Availability

Information systems and services shall remain available whenever needed for business operations.

3.5 Roles and Responsibilities

Management

- Approve security policies.
- Allocate resources.
- Support security initiatives.

IT Department

- Implement security controls.
- Monitor systems.
- Manage backups and updates.

Employees

- Follow security procedures.
- Protect passwords.
- Report security incidents.

3.6 Access Control Policy

The company shall implement Role-Based Access Control (RBAC).

Users shall receive access according to job responsibilities.

User accounts shall be reviewed every six months.

Password sharing is prohibited.

3.7 Data Protection Policy

Sensitive information shall be encrypted.

Regular backups shall be performed daily.

Data retention and secure disposal procedures shall be implemented.

3.8 Security Incident Management

All security incidents shall be reported immediately.

Incident response procedures shall include:

- Detection

- Analysis
- Containment
- Recovery
- Post-incident review

3.9 Compliance and Enforcement

Violations of security policies may result in:

- Warning
- Suspension of access privileges
- Termination of employment
- Legal action

4.0 PROGRAM POLICY

4.1 Security Governance Policy

The company shall establish an Information Security Committee responsible for:

- Security planning
- Policy review
- Compliance monitoring
- Risk management

4.2 Security Awareness and Training Policy

Employees shall receive annual information security training covering:

- Cybersecurity threats
- Password security
- Social engineering attacks
- Data protection
- Incident reporting

4.3 Risk Management Policy

The organization shall conduct risk assessments every six months.

Risk management activities shall include:

- Asset identification
- Threat assessment
- Vulnerability analysis
- Risk evaluation
- Risk treatment

4.4 Business Continuity and Disaster Recovery Policy

Business continuity plans shall be developed and maintained.

Disaster recovery procedures shall ensure restoration of critical systems within acceptable timeframes.

Backups shall be tested quarterly.

4.5 Security Auditing and Monitoring Policy

Security audits shall be conducted every six months.

Audit activities shall include:

- Access reviews
- Log analysis
- Compliance assessments
- Vulnerability assessments

5.0 ISSUE-SPECIFIC POLICY

5.1 Password Management Policy

Purpose

To protect organizational systems against unauthorized access.

Requirements

- Minimum 12 characters.
- Uppercase and lowercase letters.

- Numbers and special characters.
- Password change every 90 days.
- MFA enabled.
- Account lockout after five failed attempts.

5.2 Email Usage Policy

Employees shall:

- Use company email for business purposes.
- Avoid opening suspicious attachments.
- Report phishing emails.

Employees shall not:

- Send confidential information without authorization.
- Use company email for illegal activities.

5.3 Internet Usage Policy

Employees shall:

- Access only authorized websites.
- Follow acceptable use guidelines.

Employees shall not:

- Download unauthorized software.
- Access malicious websites.
- Engage in illegal online activities.

6.0 SYSTEM-SPECIFIC POLICY

6.1 Human Resource Module Security Policy

- Access limited to HR staff.
- Employee records encrypted.
- Payroll information protected.
- All activities logged.

6.2 Finance Module Security Policy

- Access limited to finance personnel.
- Financial transactions audited.
- Segregation of duties implemented.
- Daily backups performed.

6.3 Procurement Module Security Policy

- Supplier information protected.
- Procurement approvals required.
- Purchasing activities logged.
- Fraud monitoring implemented.

6.4 Sales Module Security Policy

- Customer information protected.
- Sales records encrypted.
- Access restricted to authorized personnel.
- Sales transactions monitored.

6.5 ERP Technical Security Controls

Authentication Controls

- Strong passwords
- MFA
- Account lockout

Network Security

- Firewalls
- Intrusion Detection Systems (IDS)
- VPN

Malware Protection

- Antivirus software
- Anti-malware solutions

System Maintenance

- Security patching
- Software updates

Backup Management

- Daily backups
- Offsite backup storage
- Recovery testing

Audit Logging

- User activity logging
- Transaction logging
- Log retention and review

7.0 CONCLUSION

Information security plays a critical role in protecting organizational information assets and ensuring business continuity. The Information Security Policy, Program Policy, Issue-Specific Policy, and System-Specific Policy developed for NextGen Business Solutions Ltd provide a comprehensive framework for managing information security risks within the ERP environment.

The implementation of these policies will help safeguard sensitive information, maintain operational efficiency, support legal compliance, and strengthen customer confidence. Continuous monitoring, employee awareness, and management commitment are essential for ensuring the effectiveness of the information security program.

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